
BIOLOGY

9744/01

Paper 1 Multiple Choice

19 September 2017
Tuesday

1 hour

Additional Materials: Multiple Choice Answer Paper

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, PDG and identification number on the Answer Sheet.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

- 1 A scientist viewing β -cells in the islets of Langerhans with a light microscope found that many of these cells contained very large nucleoli.

Which of these organelles would be found in large quantities in the cytoplasm if these cells were viewed with an electron microscope?

1. Mitochondria
2. Rough endoplasmic reticulum
3. Vesicles

- A 1, 2 and 3
B 1 and 2 only
C 1 and 3 only
D 2 and 3 only

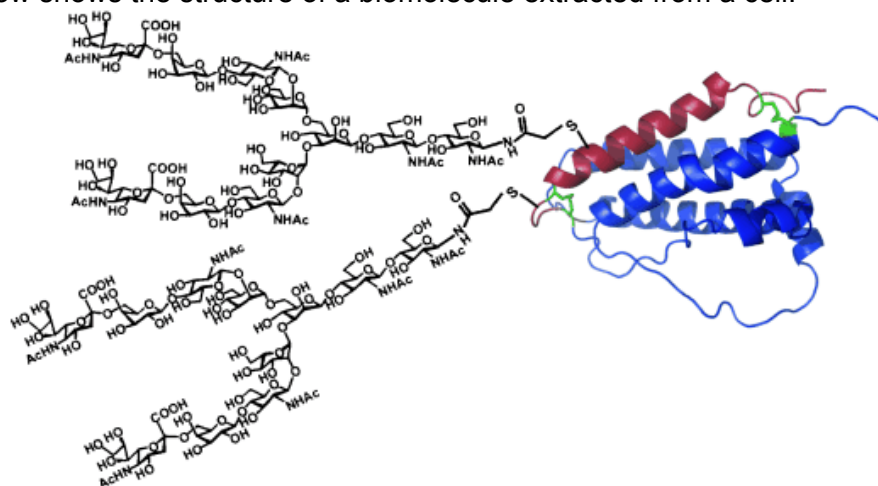
- 2 Which of the following regarding embryonic stem cells and blood stem cells is true?

- A As embryonic stem cells develop, they turned into blood stem cells as they lose their ability to differentiate into all types of cells.
B Embryonic stem cells have more genes than blood stem cells and thus are able to form more types of cells.
C Under normal conditions, embryonic stem cells express more of their genes compared to the blood stem cells.
D Both stem cells are derived from the zygotic stem cells with the blood stem cells having a lowered telomerase activity compared to the embryonic stem cells.

- 3 Which of the following options correctly matches the functional and structural features of cellulose, collagen, glycogen and triglycerides?

		Function	Structure		
			Fibrous	Molecule held together by hydrogen bonds	Branched chains
A	Cellulose	Support	✓	x	✓
	Collagen	Strengthening	✓	✓	x
B	Cellulose	Support	✓	✓	x
	Triglyceride	Energy source	x	x	x
C	Collagen	Strengthening	✓	✓	✓
	Glycogen	Storage	x	x	✓
D	Glycogen	Storage	x	✓	✓
	Triglyceride	Energy source	x	✓	x

- 4 The figure below shows the structure of a biomolecule extracted from a cell.



Below are some statements regarding the structure, property and function of biomolecules with structures similar to that shown above. Which of the following statements are true?

- 1 This biomolecule has both hydrophilic and hydrophobic properties.
- 2 This kind of biomolecule plays a role in blood group determination.
- 3 This biomolecule is contained within the secretory vesicle.
- 4 When completely hydrolysed, all the monomers of this biomolecule are soluble in water.

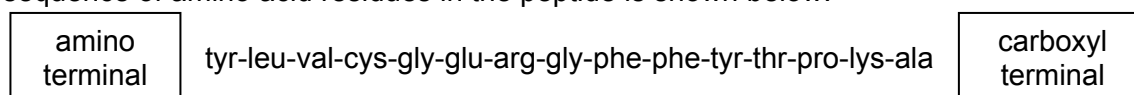
- A 1 and 3 only
 B 2 and 4 only
 C 1, 2 and 4 only
 D 2, 3 and 4 only

- 5 A peptide section of an insulin molecule was hydrolysed by two proteases, trypsin and chymotrypsin.
- Trypsin breaks the peptide bonds at the carboxyl terminals of lysine (lys) and arginine (arg).
 - Chymotrypsin breaks the peptide bonds at the carboxyl terminals of phenylalanine (phe), tryptophan (trp) and tyrosine (tyr).

The hydrolysis was performed separately using:

- (i) Both enzymes, or
 (ii) Trypsin only, or
 (iii) Chymotrypsin only.

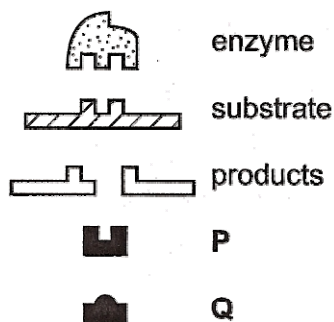
The sequence of amino acid residues in the peptide is shown below:



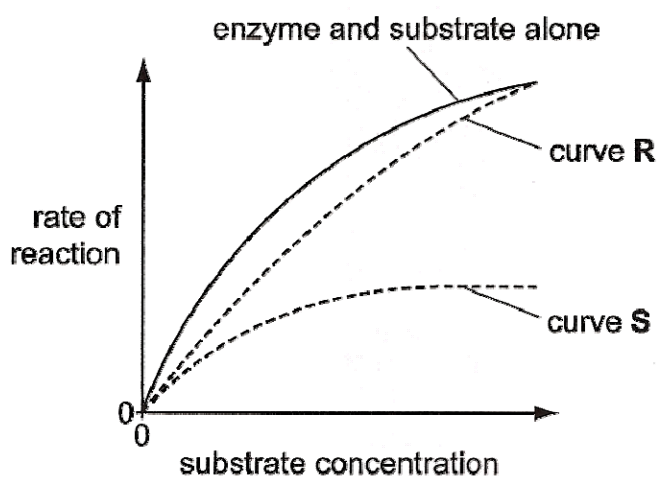
Which statement concerning the products of hydrolysis is correct?

- A Fewer than half of the fragments from hydrolysis (i) are single amino acids.
 B Hydrolysis (ii) yields one few fragment than hydrolysis (iii)
 C Hydrolysis (ii) yields one more dipeptide than hydrolysis (iii)
 D With hydrolysis (i), all fragments formed are seven or fewer amino acid residues long.

- 6 The diagram shows an enzyme molecule with its normal substrate and products. P and Q are other molecules that can bind to the enzyme.



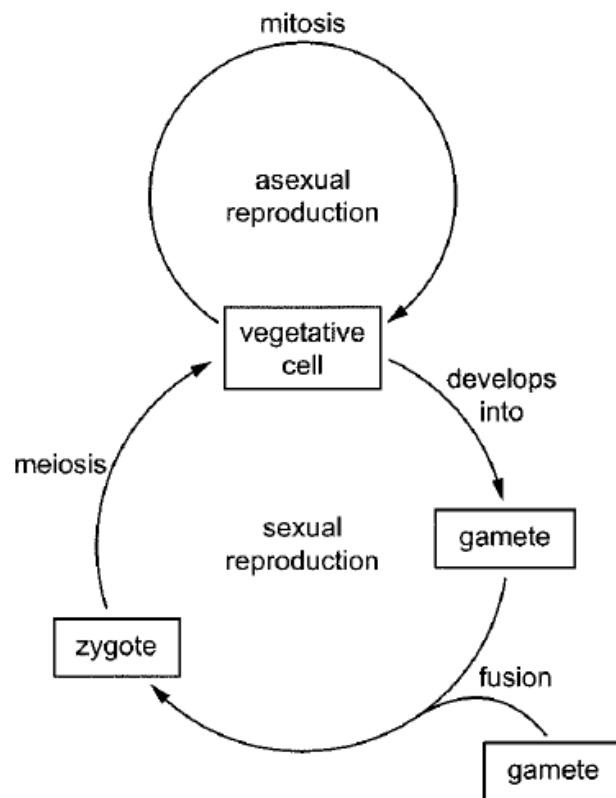
The graph shows the effect of P and Q on the rate of reaction of the enzyme at different substrate concentrations.



Which statement correctly describes the activity of the enzyme?

- A P is a competitive inhibitor that binds to the active site, resulting in curve R.
- B P is a non-competitive inhibitor that distorts the shape of the enzyme, resulting in curve S.
- C Q is a competitive inhibitor that distorts the shape of the enzyme, resulting in curve R.
- D Q is a non-competitive inhibitor that binds to the active site, resulting in curve S.

- 7 *Chlamydomonas* is a small, unicellular, green alga that undergoes asexual and sexual reproduction as part of its life cycle, as shown in the diagram.



What can be deduced from this information?

- A Fusion of gametes restores the diploid number of the vegetative cell.
- B Gametes for sexual reproduction are always formed as the products of meiosis.
- C Vegetative cells from haploid daughter cells in asexual reproduction.
- D Vegetative cells undergo meiosis to form gametes for sexual reproduction.

8 The mechanism of action of four drugs that inhibit DNA replication is stated below.

- Aphidicholine inhibits DNA polymerase
- Cytarabine is converted into a molecule that can substitute for a DNA nucleotide and also inhibits DNA repair mechanisms
- Epirubicin inhibits an enzyme involved in the unwinding of DNA and separation of strands
- Hydroxycarbamide inhibits an enzyme involved in the production of deoxyribonucleotides

Which row correctly matches a drug to an explanation of the mechanism of action?

	explanation of mechanism of action			
	decreased pool of available nucleotides inhibits chain elongation	DNA strands not available as templates for transcription	DNA damaged during replication and cell death occurs	exposed DNA template strands unable to be copied
A	aphidicholine	epirubicin	cytarabine	hydroxycarbamide
B	epirubicin	cytarabine	hydroxycarbamide	aphidicholine
C	hydroxycarbamide	aphidicholine	epirubicin	cytarabine
D	hydroxycarbamide	epirubicin	cytarabine	aphidicholine

9 A bacterium produces a normal protein with the following amino acid sequence:

Met – Val – His – Lys – Arg – Thr – Leu - Val

After irradiation, a mutant strain is produced that synthesises a mutant protein from the same coding region on DNA with the following sequence:

Met – Val – His – Lys – Glu – Pro

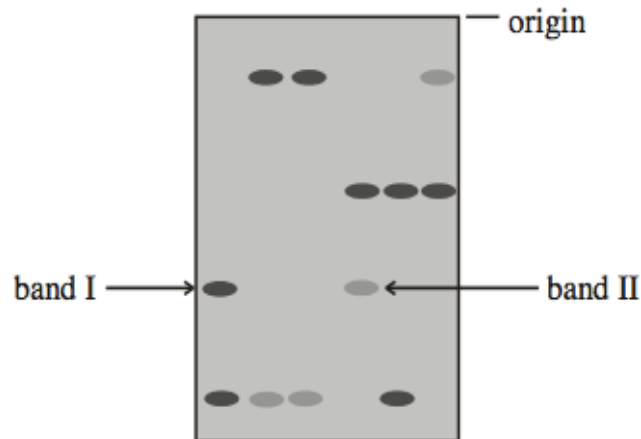
The mRNA codons for some amino acids are shown as follows:

Arg	Glu	Leu	Pro	Thr	Val
AGA	GAA GAG	UUA	CCU CCC	ACC ACG ACA	GUU GUA GUG

Which of the following mutations have occurred in the template DNA strand encoding the protein?

- A** Substitution of T by A at the 13th nucleotide position.
- B** Deletion of T at the 13th nucleotide position.
- C** Insertion of C at the 13th nucleotide position.
- D** Substitution of A by T at the 20th nucleotide position.

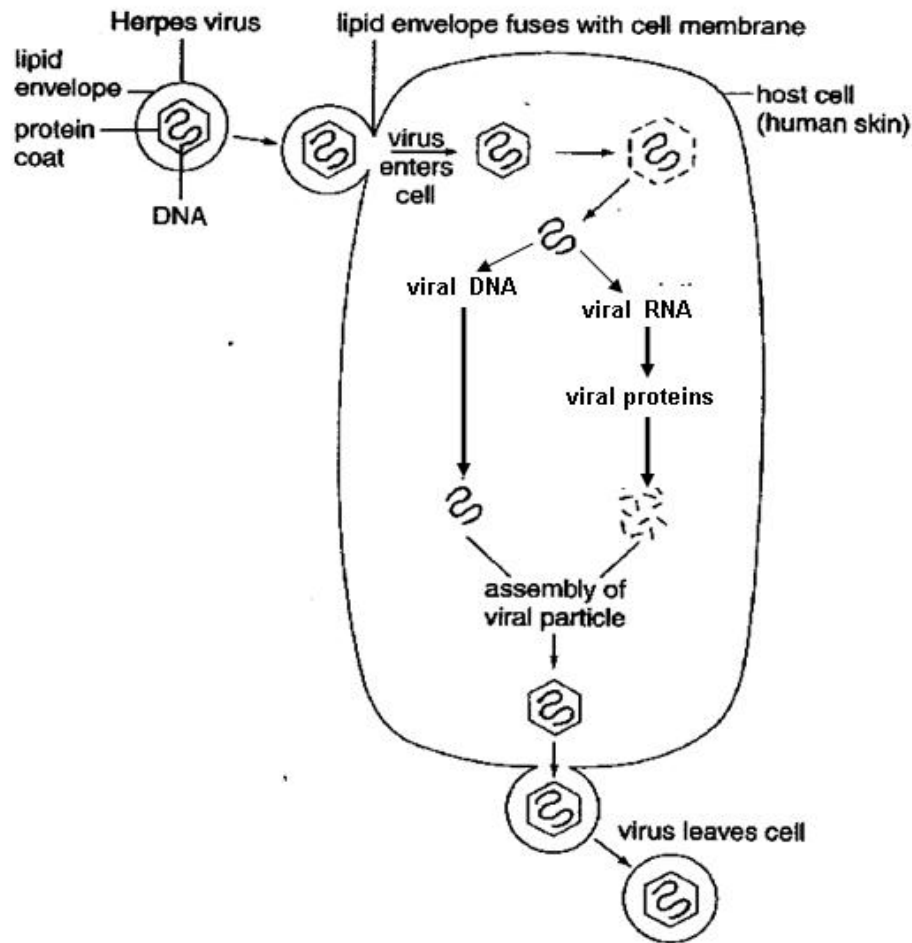
- 10 The diagram shows the results of DNA profiling using gel electrophoresis.



What conclusion can be drawn about the DNA in bands I and II?

- A The DNA in the two bands had the same base sequence.
 - B The DNA in the two bands had the same ratio of bases.
 - C The DNA in the two bands came from the same source.
 - D The DNA in the two bands have the same charge to mass ratio.
- 11 Which of the following statement comparing the human immunodeficiency virus (HIV) and lambda phage is **incorrect**?
- A The HIV enters by receptor-mediated endocytosis, but the lambda phage infects bacterial cells by injecting its DNA.
 - B The capsid of the HIV enters the host cell, but the capsid of the lambda virus does not.
 - C The genome of the HIV must be processed before it is integrated into the host chromosome, but the genome of the lambda virus can be directly integrated.
 - D New HIV are released from the host cell via budding, but new lambda virus are released via cell lysis.

- 12 The diagram below shows the reproductive cycle of the herpes virus which causes cold sores on the mouth. With reference to the diagram below, which of the following statements best describes the herpes virus?



- A It is not a retrovirus as it does not contain RNA as its genetic material
- B Its mode of replication is similar to that of influenza virus.
- C Its replication cycle includes a lysogenic phase.
- D It carries its own enzymes and ribosomes to make viral proteins.
- 13 What are the correct characteristics for a prokaryotic genome?

	Promoters	DNA always bound to histone proteins	Plasmids often present	Repeat sequences absent or uncommon
A	✓	✗	✗	✗
B	✗	✓	✓	✓
C	✗	✓	✗	✗
D	✓	✗	✓	✓

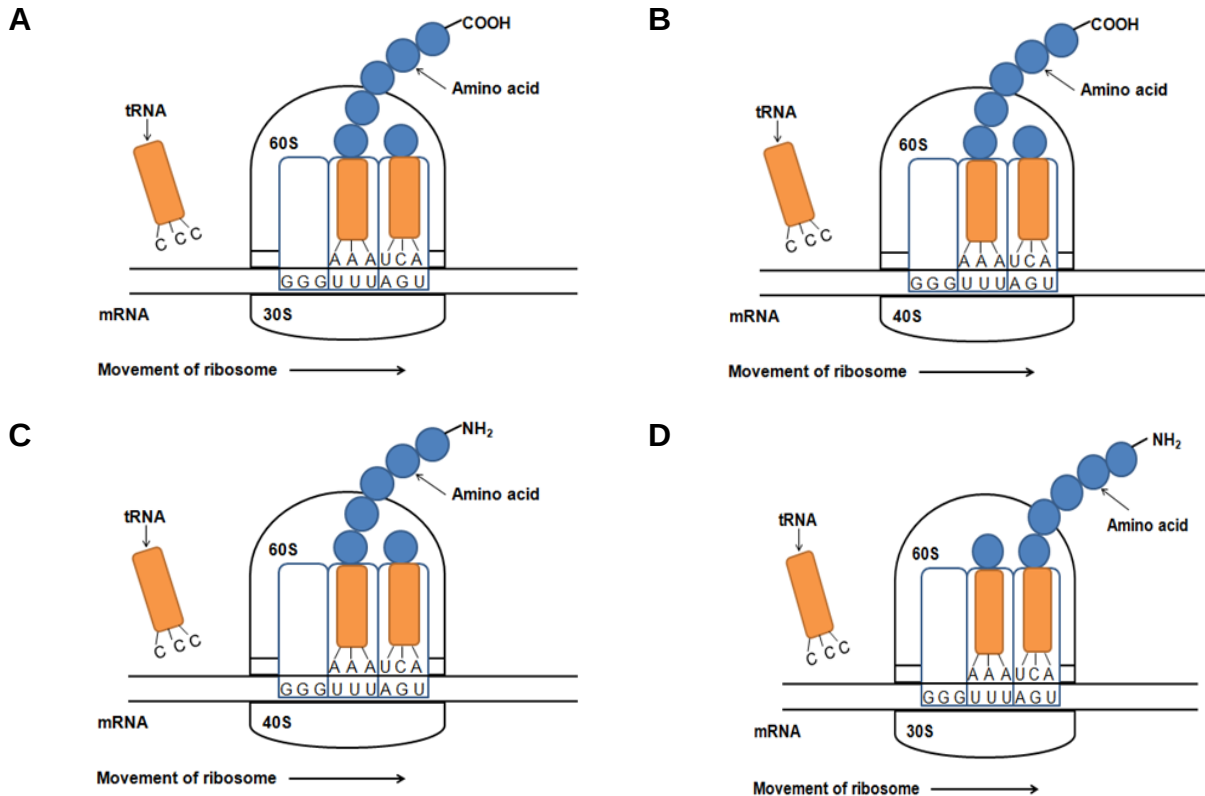
- 14** A mutant strain of *E. coli* has been isolated in which the *lac* operon is not expressed in the presence of lactose. This mutant strain was mated so that it now contains an F plasmid containing a normal *lac* operon. The mutant and mated strain with regard to their β -galactosidase activities in the presence and absence of lactose was compared. The following results were obtained:

Strain	Addition of lactose	Amount of β -galactosidase (percentage of parent strain)
Parent	No	0
Parent	Yes	0
Mated	No	0
Mated	Yes	100

With respect to the results shown in table, which part of the bacterial DNA most likely is mutated?

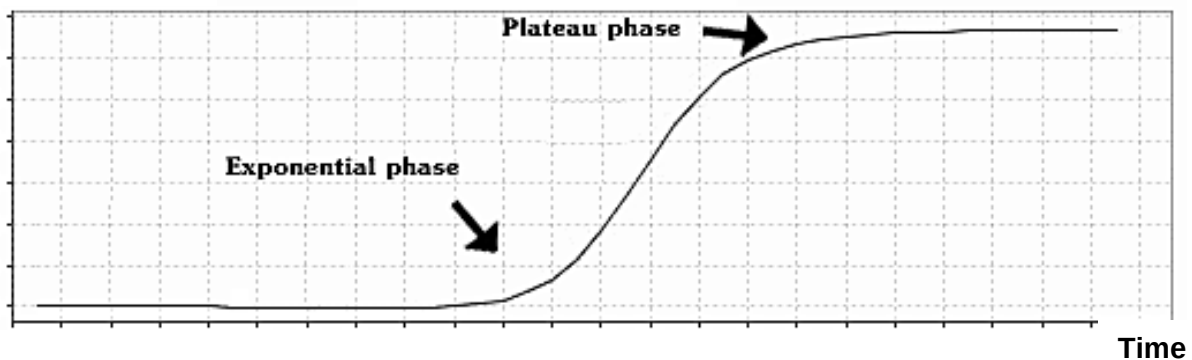
- A** *lac A*
 - B** *lac I*
 - C** Promoter of *lac* operon
 - D** Operator of *lac* operon
- 15** Which comparative statements about prokaryotic and eukaryotic gene expression are correct?
- 1 DNA methylation is a feature of prokaryotes but not eukaryotes.
 - 2 Eukaryotes and prokaryotes both use ribosomes to translate mRNA.
 - 3 Eukaryotes have introns, most prokaryotes do not.
 - 4 Prokaryotes have genes organized into operons, most eukaryotes do not.
- A** 1, 2 and 3
 - B** 1, 2 and 4
 - C** 1, 3 and 4
 - D** 2, 3 and 4

16 Which of the following diagrams shows the correct process of eukaryotic translation?



17 During PCR, the amount of DNA synthesised can be traced using fluorescent primers and the measurements are shown in the following plot. The process initially goes through an exponential phase, followed by a plateau phase eventually.

Amount of DNA



Which of the following statements is **true**?

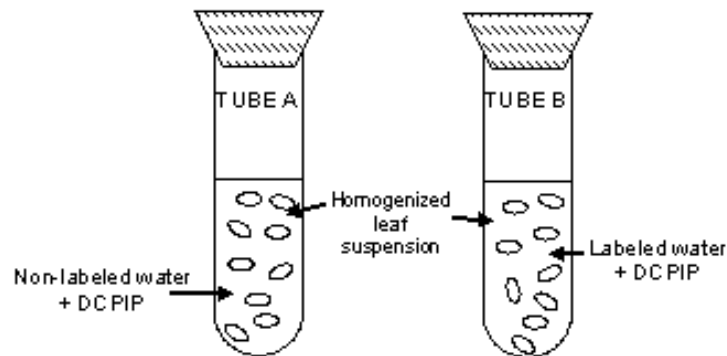
- A During the exponential phase, the number of DNA molecules synthesized after 15 cycles is 15^2 .
- B During the exponential phase, the temperature is always maintained at the optimum temperature of 72°C hence there is rapid amplification.
- C During the plateau phase, the reaction mixture is being depleted of ribonucleotides.
- D During the plateau phase, *Taq* polymerase may be denatured.

18 Which of the following is true about both cyclic and non-cyclic photophosphorylation?

- 1 Establishes an electrochemical gradient across the thylakoid membrane
- 2 Involve photosystem II
- 3 Require oxygen as the final electron acceptor
- 4 Photolysis of water occurs

- A 1 only
 B 1 and 2 only
 C 2 and 4 only
 D 1, 3 and 4 only

19 The experimental setup below was created by homogenizing leaf cells to break their cell walls. The leaf suspensions containing the cytoplasm and organelles were then placed in test-tubes containing non-labeled water (H_2^{16}O) and ^{18}O -labeled water (H_2^{18}O) respectively. A few drops of DCPIP, a hydrogen acceptor, were added to each test-tube. DCPIP will turn from blue to colourless when it is reduced and this colourless DCPIP can be reoxidized to blue.



The test-tubes were then exposed to blue light for 30 minutes. Which of the following shows the results of the two test-tubes after 30 minutes?

	Tube A		Tube B	
	Gas evolved	DCPIP colour	Gas evolved	DCPIP colour
A	C^{16}O_2	Blue	C^{18}O_2	Blue
B	$^{16}\text{O}_2$	Blue	$^{16}\text{O}_2$	Blue
C	C^{16}O_2	Colourless	C^{18}O_2	Colourless
D	$^{16}\text{O}_2$	Colourless	$^{18}\text{O}_2$	Colourless

20 From which substrate is the first carbon dioxide molecule released during cellular respiration?

- A Glucose
 B Pyruvate
 C Acetyl-coA
 D Citrate

- 21** Four tubes containing preparations from animal tissue were set up as shown in the table.

Tube	Contents
1	Glucose + homogenized cells
2	Glucose + cytoplasm lacking organelles
3	Pyruvic acid + homogenized cells
4	Pyruvic acid + mitochondria

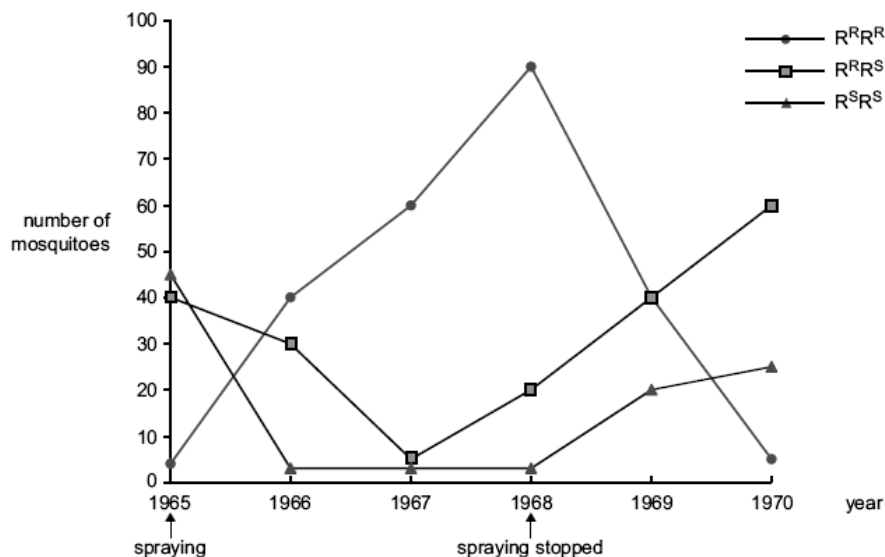
After incubation, in which tube/ tubes would at least 36 ATP be produced?

- A** 1 only
- B** 1 and 3 only
- C** 1, 2 and 4 only
- D** 1, 3 and 4 only
- 22** A plant is known to be heterozygous at two gene loci, X and Y. The pollen grains from this plant are used to fertilise another plant of the same genotype. What is the probability that an embryo will be homozygous dominant at one locus.
- A** 1 in 4
- B** 3 in 8
- C** 5 in 8
- D** 1 in 16
- 23** Barring in chickens is due to a sex-linked dominant gene. The sex of chicks at hatching is difficult to determine but barred chicks can be distinguished from nonbarred at that time. What cross would you make such that all chicks of one sex are barred?
In chicken, the male is the homogametic sex.
- A** Barred males x barred females
- B** Barred males x nonbarred females.
- C** Nonbarred males x barred females
- D** Nonbarred males x nonbarred females

- 24 Birds, such as cockatoos, have a species of louse (an insect parasite) that lives on their feathers. White sulfur-crested cockatoos have pale lice on their wings and bodies while yellow-tailed black cockatoos have dark lice on their wings and bodies. Both of these cockatoos have black lice of this species on their heads. In order to rid themselves of these parasites, cockatoos preen their wings and bodies with their beaks but have to use their feet to preen their heads.

What best explains how this species of louse has diversified into two colour variants on the birds' wings and bodies, but has remained dark on the birds' heads?

- A Cockatoo beak preening results in selection pressure on wing and body lice.
 - B Cockatoos are unable to see the lice while preening their heads.
 - C Cockatoos notice badly camouflaged lice on their wings and bodies while preening.
 - D Cockatoos use different preening techniques on different parts of their bodies resulting in natural selection.
- 25 In the mosquito, there is a gene locus which has two alleles, R^R and R^S , involved in resistance to the insecticide DDT. R^R represents the allele for DDT resistance and R^S represents the allele for DDT sensitivity. The graph shows the number of mosquitoes of three genotypes collected from 1965, when DDT was first used, through to 1970, two years after the spraying of DDT stopped.



From the data, it is possible to conclude that

- A the frequency of the R^S allele is greater than the frequency of the R^R allele in 1968.
- B many generations after the removal of DDT, the R^R allele would disappear from the population.
- C after removal of DDT from the environment in 1968, having the $R^R R^R$ genotype reduces the chance of survival.
- D in the presence of DDT in the environment between 1967 and 1968, mosquitoes with the $R^R R^S$ genotype are most likely to survive.

26 Which of the following shows the correct sequence of events?

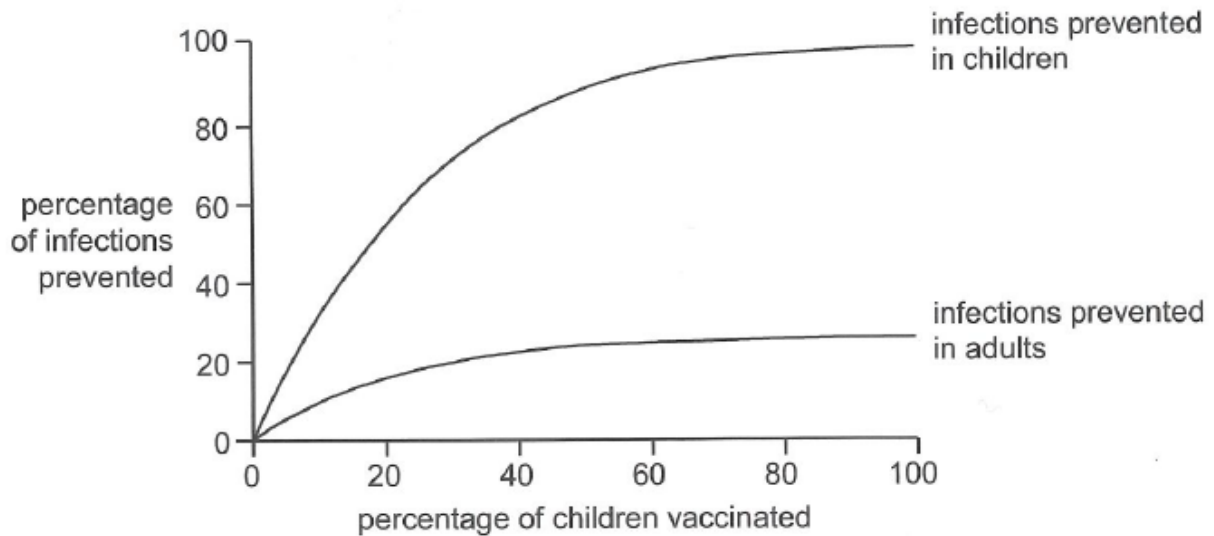
1	adaptation of a population	→	competition and predation leading to natural selection	→	behavioural isolation	→	allopatric speciation
2	adaptation of a population	→	competition and predation leading to natural selection	→	physiological isolation	→	allopatric speciation
3	competition and predation leading to natural selection	→	physiological isolation	→	adaptation of isolated populations	→	sympatric speciation
4	competition and predation leading to natural selection	→	geographical isolation	→	adaptation of isolated populations	→	allopatric speciation

- A** 3 only
- B** 1 and 2 only
- C** 3 and 4 only
- D** 1, 3 and 4 only

27 Which of the following statements regarding a B cell expressing both IgM and IgD on its membrane is incorrect?

- A** The L chains of the IgM and IgD have identical amino acid sequences.
- B** The constant parts of the H chains of the IgM and IgD have different amino acid sequences.
- C** The IgM and IgD have different antigenic specificities.
- D** If it is triggered by antigen and T-cell signals to proliferate and differentiate, it may differentiate into a plasma cell that may secrete IgG, IgE, or IgA antibodies.

28 The diagram shows the effect of vaccination of children on the prevention of infection.



What can be concluded about the effect of vaccination of children from this data?

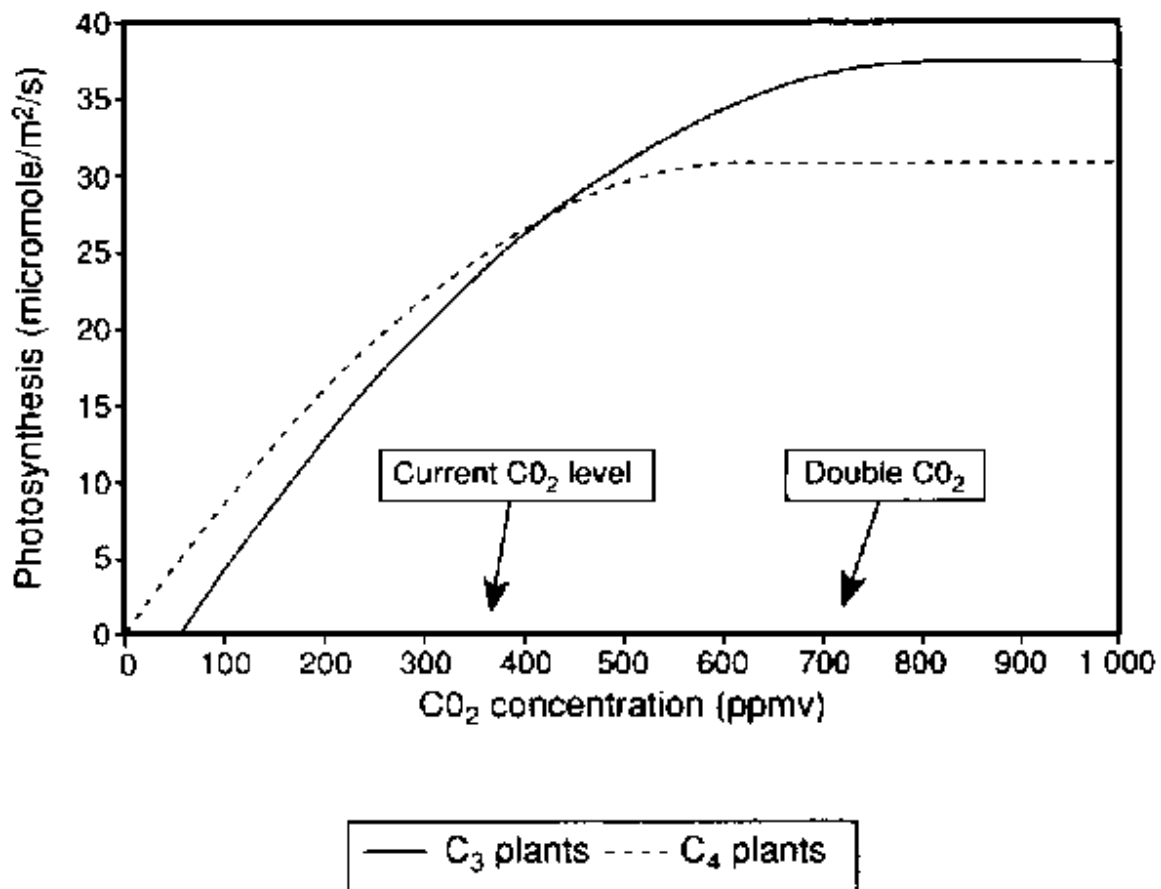
1. When approximately 80% of children are vaccinated, the cycle of disease transmission in children is broken.
2. Vaccination of children reduces the percentage of infections in both adults and children.
3. The effect on adult infection is less than that on infection in children, because adults will have been vaccinated as children
4. The effect on children infection is less than that on infection in adults, because more children are not suitable candidates for vaccination.

- A 1 and 2 only
 B 1 and 4 only
 C 2 and 3 only
 D 1, 2 and 4

29 What contributes to the enhanced greenhouse effect?

- A Ozone from violent thunderstorms
 B Carbon particles in diesel engine exhaust
 C Methane from agricultural sources
 D Carbon dioxide from active volcanoes around the world

- 30 The graph shows the impact of climate change on C₃ and C₄ plants. C₃ and C₄ are the different types of photosynthetic methods used by different plants.



Which of the following statement is supported by the graph?

- A Both C₃ and C₄ plants photosynthetic rate are only limited by carbon dioxide concentration currently.
- B C₃ and C₄ plants react to environmental stress caused by an increase in carbon dioxide differently.
- C Farmers may want to consider growing more C₃ crop plants in future
- D Global warming increases the yield of crop plants for both cattle and human.

Answers

1	A	16	C
2	D	17	D
3	B	18	A
4	C	19	D
5	D	20	B
6	A	21	A
7	C	22	B
8	D	23	C
9	B	24	D
10	D	25	C
11	A	26	C
12	A	27	C
13	D	28	A
14	C	29	C
15	D	30	C